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HTML MEDIA – AUDIO & VIDEO

HTML `<audio>` and `<video>` tags are used to embed sound and video files in web pages and play them in the browser.

AUDIO

METHOD 1: DIRECT SRC METHOD (SINGLE FILE)

```html
<audio src="audio.mp3" controls>
</audio>
```

- controls attribute adds audio controls (play, pause, volume, time).
- Use this method when you have only one audio file.

REAL WORLD OUTPUT EXAMPLE

METHOD 2: MULTIPLE AUDIO SOURCES (BACKUP)

```html
<audio controls>
  <source src="audio.mp3" type="audio/mpeg">
  <source src="audio.ogg" type="audio/ogg">
  <source src="audio.wav" type="audio/wav">

  <p>Your browser does not support the audio tag.</p>
</audio>
```

Browser tries the first <source> file.
↓
If that format is not supported, it tries the next <source> file.
↓
If none of the formats are supported, the message inside <p> is displayed.

REAL WORLD OUTPUT EXAMPLE

FALLBACK MESSAGE (SHOWN ONLY IF NO FORMAT IS SUPPORTED)
Your browser does not support the audio tag.

IMPORTANT AUDIO POINTS

- ✓ controls attribute shows audio player controls (play, pause, volume, time).
- ✓ src attribute defines the path of the audio file.
- ✓ Multiple <source> elements improve compatibility.
- ✓ Supported formats: MP3, OGG, WAV, AAC

AUDIO USE CASES (REAL WORLD)

- Podcasts
- Online Music Players
- Language Pronunciation
- E-learning Lectures

AUDIO BENEFITS

- Better Compatibility
- Smooth User Experience
- No Errors on Unsupported Browsers
- Professional Websites

VIDEO

METHOD 1: DIRECT SRC METHOD (SINGLE FILE)

```html
<video src="video.mp4" controls
       width="600" height="350">
</video>
```

- controls → shows video controls.
- width → sets the video width.
- height → sets the video height.
- Supported formats: MP4, WebM, OGG

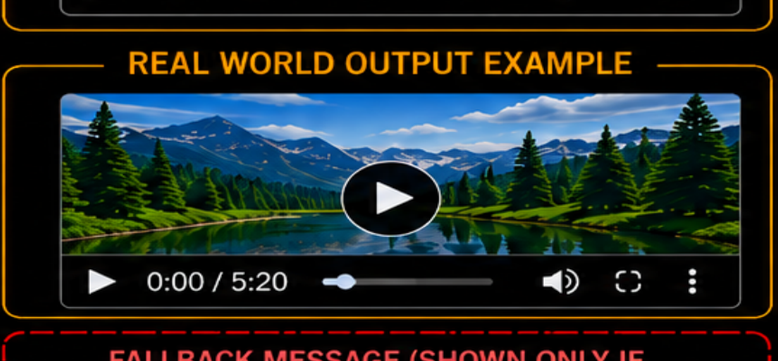
REAL WORLD OUTPUT EXAMPLE

METHOD 2: MULTIPLE VIDEO SOURCES (BACKUP)

```html
<video controls width="600" height="350">
  <source src="video.mp4" type="video/mp4">
  <source src="video.webm" type="video/webm">
  <source src="video.ogv" type="video/ogg">
  <p>Your browser does not support the video tag.</p>
</video>
```

Browser tries the first <source> file.
↓
If that format is not supported, it tries the next <source> file.
↓
If none of the formats are supported, the message inside <p> is displayed.

REAL WORLD OUTPUT EXAMPLE



FALLBACK MESSAGE (SHOWN ONLY IF NO FORMAT IS SUPPORTED)
Your browser does not support the video tag.

VIDEO USE CASES (REAL WORLD)

- Online Courses
- Movie Streaming
- Product Demos
- News Videos
- Social Media Videos

AUTOPLAY VIDEO (USE WITH CAUTION)

```html
<video controls autoplay muted loop
 width="600" height="350">
  <source src="video.mp4" type="video/mp4">
</video>
```

- autoplay → video starts automatically.
- muted → required for autoplay (most browsers).
- loop → video repeats continuously.

Note: Autoplay with sound is usually blocked in most browsers.
Use muted to avoid this issue.

IMPORTANT NOTES

1. Browser supports the first supported format in play order.
2. If no format is supported, fallback text (<p>) is shown.
3. Do not use direct src and <source> together.
4. Use multiple <source> elements for better compatibility.
5. Provide the correct type attribute in each <source>.

BEST PRACTICES

- ☑ Use multiple formats for compatibility.
- ☑ Optimize file size (compression).
- ☑ Use real compression for fast loading.
- ☑ Ensure accessibility with controls.
- ☑ Always provide fallback text inside <p>.
- ☑ Use autoplay only when necessary.

FILE STORAGE TIPS

Store audio/video files in:
- ✓ Your Server
- ✓ Cloud Storage (AWS S3, GCP, Azure)
- ✓ CDN (Cloudflare, AWS CloudFront)

Use optimized formats for faster loading and better user experience.

SUPPORTED FORMATS (QUICK REFERENCE)

AUDIO – Best compatibility
OGG – Open source, good quality
WAV – Uncompressed (big size)
AAC – Better quality (modern)

MP4 – Best compatibility
WebM – Modern, optimized
OGG (Ogv) – Open source
(Not fully supported in all browsers)

WHERE TO STORE MEDIA FILES?

- Own Server
- Cloud Storage
- CDN

BENEFITS OF GOOD STORAGE

- ✓ Fast loading speed
- ✓ High availability
- ✓ Better user experience
- ✓ Low bandwidth cost

YOUTUBE VIDEO – HOW TO ATTACH & PLAY

1 WHAT IS <IFRAME>?

The <iframe> tag is used to embed another HTML document inside the current document.

2 SYNTAX

```html
<iframe src="URL" width="WIDTH" height="HEIGHT"
        frameborder="0" allow="ALLOW_FEATURES"
        allowfullscreen></iframe>
```

IFRAME PREVIEW

3 YOUTUBE VIDEO – HOW TO ATTACH & PLAY

STEP 1: GET THE VIDEO LINK

Go to YouTube and copy the video URL.

https://www.youtube.com/watch?v=AuKZ3WdCUy4

STEP 2: CONVERT WATCH LINK TO EMBED LINK

Replace watch?v=VIDEO_ID with embed/VIDEO_ID

Watch Link (URL)
https://www.youtube.com/watch?v=AuKZ3WdCUy4
↓
Embed Link (for iframe src)
https://www.youtube.com/embed/AuKZ3WdCUy4

STEP 3: USE <IFRAME> TO EMBED THE VIDEO

```html
<iframe src="https://www.youtube.com/embed/AuKZ3WdCUy4"
 frameborder="0" allow="allowfullscreen" allowfullscreen></iframe>
```

OUTPUT (YOUTUBE VIDEO)

4 START VIDEO FROM A SPECIFIC TIME (SEEK)

Add start time using ?start=SECONDS (or &start=SECONDS)
Example: Start from 1 minute 30 seconds (90 seconds)

```html
<iframe src="https://www.youtube.com/embed/AuKZ3WdCUy4?start=90"
  frameborder="0" allow="allowfullscreen" allowfullscreen></iframe>
```

More Examples
- Start from 10 seconds → ?start=10
- Start from 2 minutes → ?start=120
- Start from 1 hour → ?start=3600
- Start from 1:30 (1 min 30 sec) → ?start=90

5 END VIDEO AT A SPECIFIC TIME

Add end time using &end=SECONDS
Example: Start from 30 sec and end at 1 min 30 sec (90 sec)

```html
<iframe src="https://www.youtube.com/embed/AuKZ3WdCUy4?start=30&end=90"
  frameborder="0" allow="allowfullscreen" allowfullscreen></iframe>
```

Note: end time must be greater than start time.

6 AUTOPLAY VIDEO (USE WITH CAUTION)

```html
<iframe src="https://www.youtube.com/embed/AuKZ3WdCUy4?autoplay=1"
  frameborder="0" allow="autoplay; encrypted-media"
  allowfullscreen></iframe>
```

- autoplay=1 → video starts automatically.
- muted=1 → required for autoplay (most browsers).
- loop=1 → video repeats continuously.
- playlist=VIDEO_ID → required with loop=1.

Note: Autoplay with sound is usually blocked in most browsers.
Use muted=1 to avoid this issue.

WHAT IS ENCRYPTED-MEDIA?

The encrypted-media permission allows the iframe to play DRM (Digital Rights Management) protected or encrypted content securely.

REAL WORLD PLATFORMS USING IT

- YouTube
- Netflix
- Amazon Prime
- Disney+
- Spotify

WHAT DOES IT DO?

- ✓ Allows playback of DRM protected videos.
- ✓ Ensures licensed content works properly.
- ✓ Helps in platforms where content is encrypted (e.g. YouTube, Netflix).
- ✓ Prevents playback issues in modern browsers.

WHAT IS DRM?

DRM is a technology used by content owners to protect their videos from unauthorized copying, downloading or redistribution.

HOW IT WORKS?

Browser ----→ Check Permission ----→ Play Protected Media

Without encrypted-media → Protected video may be blocked or restricted.

IMPORTANT NOTE

YouTube adds encrypted-media automatically because some videos have licensed, encrypted or protected streams.

Even if your video is normal, it ensures future compatibility and smooth playback.

7 LOOP VIDEO

```html
<iframe src="https://www.youtube.com/embed/AuKZ3WdCUy4?loop=1&playlist=AuKZ3WdCUy4"
  frameborder="0" allow="autoplay; encrypted-media"
  allowfullscreen></iframe>
```

Important:
- Use loop=1 to loop the video.
- Use playlist=VIDEO_ID (same as video id) for looping.

WHY playlist=VIDEO_ID IS REQUIRED?

YouTube treats loop with playlist.
For single video looping, you must pass the same VIDEO_ID in playlist.

Video Ends → No playlist specified → Loop may fail / Stops

Video Ends → Playlist has same VIDEO_ID → Video plays again (Loop works) ✓

EXAMPLES

☒ Wrong / Incomplete (may not loop properly)
https://www.youtube.com/embed/VIDEO_ID?loop=1

✓ Correct (proper looping)
https://www.youtube.com/embed/VIDEO_ID?loop=1&playlist=VIDEO_ID

MULTIPLE VIDEO LOOPING

Looping multiple videos in a playlist.
https://www.youtube.com/embed/VIDEO_ID?loop=1&playlist=id1,id2,id3

Plays id1 → id2 → id3 → repeat

HOW YOUTUBE LOOP WORKS INTERNALLY

Without playlist
Loop=1 Enabled → No playlist specified → No replay target → Stops

With playlist
Loop=1 Enabled → Playlist Specified → Replay starts again → Continuous Loop

COMMON ALLOW PERMISSIONS

- autoplay – Auto play allowed
- encrypted-media – Protected media allowed (DRM)
- fullscreen – Full screen allowed
- picture-in-picture – Mini floating player allowed

KEY TAKEAWAY

- encrypted-media → required for protected content.
- loop=1 → enables looping.
- playlist=VIDEO_ID → required for proper looping (single video).
- autoplay + muted → needed for auto play in most browsers.

8 ALL POSSIBLE REAL-WORLD USE CASES OF <IFRAME> (YOUTUBE VIDEO)

- Embed YouTube videos in blogs & articles
- Show product demo videos on e-commerce sites
- Add tutorial & how-to videos in educational websites
- Embed videos in news & magazine websites
- Use in portfolios to show project videos
- Embed videos in landing pages
- Use in LMS / e-learning platforms
- Add videos in online courses
- Embed videos in intranet systems
- Show promotional videos on business websites
- Embed viveot live streams (past replays)
- Use in help centers & knowledge base
- Embed in forums & community sites
- Add videos in real estate property listings
- Embed in dashboards (iframe widgets)

9 OTHER IFRAME TIPS

- ✓ Always use HTTPS URLs for security.
- ✓ Use allowfullscreen for better user experience.
- ✓ Set appropriate width and height.
- ✓ Use responsive design for mobile devices.
- ✓ Be careful with third-party content for performance.

10 SAMPLE RESPONSIVE IFRAME (YOUTUBE)

```html
<div style="position: relative; padding-bottom: 56.25%;
        height: 0; overflow: hidden;max-width: 100%;">
  <iframe src="https://www.youtube.com/embed/AuKZ3WdCUy4"
    style="position: absolute; top: 0; left: 0; width: 100%;
    height; 100%; border: 0;" allow="allowfullscreen"
    allowfullscreen></iframe>
</div>
```

BENEFITS OF USING <IFRAME>

- Easy to implement
- Improved User Experience
- Saves Development Time
- Highly Flexible
- Supports Multiple Content Types
- Enhances Website Functionality